

Annex A Mapping to ISA/IEC 62443-4-2 (informative)



This Annex provides a mapping of [OPC Profiles – OPC UA Profiles and Facets](#) UA Profiles and Facets

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ISA/IEC 62443-4-2 to OPC UA. The first 5 columns (yellow colour) in each table are from the [OPC Profiles – OPC UA Profiles and Facets](#) UA Profiles and Facets

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ISA/IEC 62443-4-2 specification and are included here for clarity. Some topics in [OPC Profiles – OPC UA Profiles and Facets](#) UA Profiles and Facets

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ISA/IEC 62443-4-2 do not apply to OPC UA and are marked as "N". [OPC Profiles – OPC UA Profiles and Facets](#) UA Profiles and Facets

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ISA/IEC 62443-4-2 topics that do apply are marked as "Y". ISA/IEC 62443-4-2 topics that are partially covered are marked as "P". For each topic that does apply the table lists the relevant OPC UA Parts and the [Profiles/ ConformanceUnits](#) that covers the listed functionality. For each OPC UA Part that is listed (other than [OPC 10000-7](#)) the text in the "Keyword text or comment" column is the text that should be searched for. The section that are found related to the keyword will describe the OPC UA related functionality. For the row labelled [OPC 10000-7 - ConformanceUnits](#), the keyword column contains the actual [ConformanceUnits](#) that apply. For the row labelled [OPC 10000-7 - Profiles](#), the keyword column contains the actual [Profiles/Facets](#) that apply. For more detail on the [ConformanceUnit](#) or [Profile](#), please use the on-line Profile Application available at [OPC Profiles](#).

The mapping only covers the core OPC UA specifications (Parts 1-26). Companion specification and other base OPC UA specifications may also apply to [OPC Profiles – OPC UA Profiles and Facets](#) UA Profiles and Facets

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ISA/IEC 62443-4-2 and could result in a more complete mapping.

Columns 2 through 4 in each of the tables could include a checkmark. The check marks when present indicate that the specific 62443

requirement applies to the listed security level (SL). This information is provided by the IEC 62443 specification and is reproduced here as a convenience.

The tables are broken into seven separate tables:

1. Identification and authentication control ([Table A.1](#))
2. Use control ([Table A.2](#))
3. System integrity ([Table A.3](#))
4. Data confidentiality ([Table A.4](#))
5. Restricted data flow ([Table A.5](#))
6. Timely response to events ([Table A.6](#))
7. Resource availability ([Table A.7](#))

ISA/IEC 62443 extends beyond just OPC UA and also requires all software be compliant. This mapping only describes how OPC UA applies, it does not describe what else would be required.

Table A.1 – ISA/IEC 62443 Mapping FR 1 Identification and authentication control

ISA/IEC 62443-4-2 CRs and Res	Related to				Applies to OPC UA	OPC UA Part #	Keyword text or comment
	SL1	SL2	SL3	SL4			
CR 1.1: Human user identification and authentication	√	√	√	√	Y	OPC 10000-4	IssuedIdentityToken
						OPC 10000-6	JSON Web Token (JWT) UserTokenPolicy
						OPC 10000-7- ConformanceUnits	Security User JWT Issue 2, Security User JWT Token Policy, OPC UA Authority Profile OAuth2 Authority Profile Azure Identity Provider Authority Profile
						OPC 10000-7 - Profiles	
CR 1.1 RE (1): Unique identification and Authentication		√	√	√	Y	OPC 10000-4	IssuedIdentityToken
						OPC 10000-6	JSON Web Token (JWT) UserTokenPolicy

					OPC 10000-7-ConformanceUnits	Security User JWT Issue 2, Security User JWT Token Policy, OPC UA Authority Profile
					OPC 10000-7 - Profiles	User Token – JWT Server User Token – JWT Client
CR1.1 RE(2) Multifactor authentication for all interfaces	√	√	P			OPC UA does not define alternate schemes for two factor authentication, but Issued tokens are used for Authentication, the issuer token provider can implement multifactor authentication. OPC UA authenticates the application as well as the
					OPC 10000-4	IssuedIdentityToken
					OPC 10000-6	JSON Web Token (JWT) UserTokenPolicy
					OPC 10000-7-ConformanceUnits	Security User JWT Issue 2, Security User JWT Token Policy, OPC UA Authority Profile
					OPC 10000-7 - Profiles	User Token – JWT Server User Token – JWT Client
CR 1.2: Software process and device identification and authentication	√	√	√	Y		ApplicationAuthentication X.509 v3 Security Certificate
					OPC 10000-4	ApplicationInstance Security Certificate
					OPC 10000-6	EndpointDescription, EndpointUrl, Hostname (Device)
					OPC 10000-7-ConformanceUnits	Security Default ApplicationInstance Certificate
					OPC 10000-7 - Profiles	Global Certificate Management Server Function Global Certificate Management Client Function

CR 1.2 RE (1) Unique identification and authentication	√	√	Y			ApplicationAuthenticati X.509 v3 Security Certifi
					OPC 10000-4	ApplicationInstance Sec Certificate
					OPC 10000-6	EndpointDescription, EndpointUrl, Hostname (Device)
					OPC 10000-7- ConformanceUnits	Security Default ApplicationInstance Cer
CR 1.3: Account management	√	√	√	P	OPC 10000-7 - Profiles	Global Certificate Management Server Fac Global Certificate Management Client Fac
						OPC UA does not direct provide account manag but if an AuthorizationS is used for user Authent it could support account management.
					OPC 10000-4	IssuedIdentityToken
					OPC 10000-6	JSON Web Token (JWT) UserTokenPolicy
CR 1.4: Identifier management	√	√	√	Y	OPC 10000-7- ConformanceUnits	Security User JWT Issue 2, Security User JWT Token Policy, OPC UA Authority Profi
					OPC 10000-7 - Profiles	User Token – JWT Serve User Token – JWT Clie
					OPC 10000-4	UserIdentityToken, UserTokenPolicy
					OPC 10000-7- ConformanceUnits	Security User JWT Issue 2, Security User JWT Token Policy, OPC UA Authority Profi
					OPC 10000-7 - Profiles	User Token – JWT Serve User Token – JWT Clie

CR 1.5: Authenticator management	√	√	√	√	Y	OPC 10000-4	UserIdentityToken, UserTokenPolicy
						OPC 10000-7- ConformanceUnits	Security User JWT Issue 2, Security User JWT Token Policy, OPC UA Authority Profi
						OPC 10000-7 - Profiles	User Token – JWT Serve User Token – JWT Client
CR 1.5 RE (1) Hardware security for authenticators		√	√		N		Secure elements are recommended in 9.1 and discussed in OPC 10000- and OPC 10000-21, but defined in OPC.
NDR 1.6 – Wireless access management	√	√	√	√	N		OPC UA does specify pl characteristics of a netw including wireless.
NDR 1.6 RE (1) Unique identification and authentication	√	√	√		N		Same as above
CR 1.7: Strength of password based authentication	√	√	√	√	N		OPC UA provides the mechanism for exchange passwords information, does not define the implementation of the password – this is vend specific. If an AuthorizationService is can provide password s enforcement.
CR 1.7 RE (1) Password generation and lifetime restrictions for human users		√	√		N		Same as CR 1.7 above.
CR 1.7 RE (2) Password lifetime			√		N		Same as CR 1.7 above.

restrictions for
all users
(human,
software
process, or
device)

CR 1.8: Security certificates	√	√	√	Y	Security Certificates, Tru (CertificateStore), OPC U Security Services
					OPC 10000-4 Obtaining, validating, a installing Security Certif services
					OPC 10000-6 Security Certificates
					OPC 10000-7- ConformanceUnits Security Certificate Administration, Security Certificate Valid
					OPC 10000-7 - Profiles Global Security Certificate Management Server Global Certificate Management Client Fac
					OPC 10000-12 Security Certificate Management Overview
CR 1.9: Strength of public key- based authentication	√	√	√	Y	Cryptographic Keys
					OPC 10000-4 Trusted Security Certific
					OPC 10000-7- ConformanceUnits Basic256_Limits,
					OPC 10000-7 - Profiles SecurityPolicy [A] – Aes128_Sha256_RsaOae SecurityPolicy [B] – Basic256Sha256 SecurityPolicy - Aes256- Sha256-RsaPss
CR 1.9 RE (1) Hardware security for public key- based authentication		√	√	N	Secure elements are recommended in 9.1 an discussed in OPC 10000 and OPC 10000-21 , but defined in OPC.
CR 1.10: Authenticator	√	√	√	Y	ApplicationAuthenticati X.509 v3 Security Certifi

feedback

OPC 10000-4	ApplicationInstance Sec Certificate
OPC 10000-6	EndpointDescription, EndpointUrl, Hostname (Device)
OPC 10000-7- ConformanceUnits	Security Default ApplicationInstance Cer
OPC 10000-7 - Profiles	Global Certificate Management Server Fac

CR 1.11: Unsuccessful login attempts	√	√	√	√	N	OPC does not provide temporary lock out for repeated user access fa but an AuthenticationSe could. OPC does monito SecureChannel connect could block secure char connection for repeated login failure.
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CR 1.12: System use notification	√	√	√	√	N	OPC does not define th a client prompts for username/password.
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NDR 1.13 – Access via untrusted networks	√	√	√	√	N	OPC does not define ne hardware requirements restrict communication between uniquely ident applications (see CR 1.2
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NDR 1.13 RE (1) Explicit access request approval		√	√	N	Same as NDR 1.13 abov
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CR 1.14: Strength of symmetric key-based authentication	√	√	√	Y	Symmetric Encryption
					OPC 10000-6 SymmetricEncryptionAl
					OPC 10000-7- ConformanceUnits
					OPC 10000-7 - Profiles Global Service KeyCred Pull Facet Global Service KeyCred Push Facet KeyCredential Service S Facet

						KeyCredential Service C Facet
					Part 14	SecuritKeyService (SKS), SymmetricEncryptionAl
CR 1.14 RE (1)	√	√		N		The OPC UA specificatio not provide hardware requirements and does utilize long lived symme keys
Hardware security for symmetric key-based authentication						

Table A.2 – ISA/IEC 62443 mapping FR 2 Use control

ISA/IEC 62443-4-2 CRs and Res	Related to				Applies to OPC UA	OPC UA Part #	Keyword text or comm
	SL1	SL2	SL3	SL4			
CR 2.1: Authorization enforcement	√	√	√	√	Y		UserAuthorization
						OPC 10000-4	Authorization Services, IssuedIdentityToken
						OPC 10000-6	AuthorizationService, JSON Web Token (JWT
						OPC 10000-7- ConformanceUnits	
						OPC 10000-7 - Profiles	User Token – JWT Serv Facet, User Token – JWT Clie Facet
RE (1): Authorization enforcement for all users (humans, software processes, and devices)	√	√	√		Y		UserAuthorization
						OPC 10000-4	Authorization Services, IssuedIdentityToken
						OPC 10000-6	AuthorizationService, JSON Web Token (JWT
						OPC 10000-7- ConformanceUnits	
						OPC 10000-7 - Profiles	User Token – JWT Serv Facet, User Token – JWT Clie Facet
RE (2): Permission	√	√	√		Y		Roles, JWT, and User R

mapping to roles						OPC 10000-18	User Authorization, Role Type
						OPC 10000-6	RolePermissions
						OPC 10000-7 - ConformanceUnits	
						OPC 10000-7 - Profiles	User Role Management 2022 Server Facet User Role Management Client Facet
CR 2.1 RE (3) Supervisor override		✓	✓		P		OPC provides the ability to switch user context, it does not provide an automatic timeout for switch.
						OPC 10000-4	Authorization Services
CR 2.1 RE (4) Dual approval				✓	N		OPC does not define the logic for applications and thus does not define dual approval
CR 2.2: Wireless use control	✓	✓	✓	✓	N		OPC is hardware agnostic
CR 2.3 – Use control for portable and mobile devices	NA	NA	NA	NA	N		No component level requirements defined in 62443
SAR 2.4: Mobile code	✓	✓	✓	✓	N		OPC does not define Mobile code technology
SAR 2.4 RE (1): Mobile code authenticity check		✓	✓	✓	N		OPC does not define Mobile code technology
EDR 2.4: Mobile code	✓	✓	✓	✓	N		OPC does not define Mobile code technology
EDR 2RE (1): Mobile code authenticity check		✓	✓	✓	N		OPC does not define Mobile code technology
HDR 2.4: Mobile code	✓	✓	✓	✓	N		OPC does not define Mobile code technology

HDR 2.4RE (1): Mobile code authenticity check	✓	✓	✓	N	OPC does not define Mobile code technolog
NDR 2.4 – Mobile code	✓	✓	✓	✓	N OPC does not define Mobile code technolog
NDR 2.4 – Mobile code	✓	✓	✓	N	OPC does not define Mobile code technolog
CR 2.5: Session lock	✓	✓	✓	✓	N OPC UA does not defin human user interface
CR 2.6: Remote session termination	✓	✓	✓	P	OPC allows identificati of remote session via ApplicationAuthenticat remote restrictions are application specific, bu the infrastructure is provided.
					OPC 10000-4 IssuedIdentityToken
					OPC 10000-6 JSON Web Token (JWT JWT UserTokenPolicy
					OPC 10000-7- ConformanceUnits Security User JWT IssuedToken 2, Security User JWT Tok Policy, OPC UA Authority Prof
					OPC 10000-7 - Profiles
CR 2.7 – Concurrent session control		✓	✓	N	OPC provides limits or sessions, subscription other functionality and defines behaviour if th limits are exceeded, bu does not provide limits per user.
CR 2.8: Auditable events	✓	✓	✓	✓	Y Auditability, Auditing, Audit Event Managem
					OPC 10000-4 Auditing
					OPC 10000-5 AuditSecurityEventTyp

						OPC 10000-7 - ConformanceUnits	
						OPC 10000-7 - Profiles	Auditing 2022 Server Facet Auditing Client Facet, Best Practice – Audit Events
CR 2.9: Audit storage capacity	√	√	√	√	N		OPC does not define Audit storage.
CR 2.9 RE (1) Warn when audit record storage capacity threshold reached.			√	√	N		OPC does not define Audit storage.
CR 2.10: Response to audit processing failures	√	√	√	√	N		OPC does not provide Audit storage
CR 2.11: Timestamps	√	√	√	√	Y		Message replay, Timestamps, SecureChannelId
						OPC 10000-4	TimestampsToReturn
						OPC 10000-5	AuditEventType
						OPC 10000-7 - ConformanceUnits	
						OPC 10000-7 - Profiles	Auditing 2022 Server Facet
CR 2.11 RE (1): Time synchronization		√	√	√	Y		Cryptographic Keys (ti validity of security pro
						OPC 10000-4	SourceTimestamp, VersionTime, Redunda Server Set Requiremen
						OPC 10000-6	Time Synchronization
						OPC 10000-7 - ConformanceUnits	

					OPC 10000-7 – Profiles	Security Time Synchronization
CR 2.11 RE (2) Protection of time source integrity			√		N	OPC does not define a unique time synchronization scheme but utilize other industry standard.
CR 2.12: Non-repudiation	√	√	√	√	P	The connection is secured against repudiation, with the exception that we use a symmetric key for the communication. Thus every message must have been signed by either the Client or the Server, but it is not possible to determine which party signed it. Message alteration, Session Hijacking, System Profiling, System Hijacking, Repudiation, Audit Event Management OPC 10000-4 Signing, GetEndpoints, SecureChannel, Audit Trail, Proof of Possession, OPC 10000-7 - Conformance Units OPC 10000-7 – Profiles User Token – JWT Server Facet, User Token – JWT Client Facet Auditing 2022 Server Facet Auditing Client Facet Best Practice – Audit Events
CR 2.12 RE (1) Non-repudiation for all users				√	P	The connection is secured against repudiation, with the exception that we use a symmetric key for the communication. Thus every message must have been signed by either

Client or the Server, but is not possible to determine which party signed it.

	Message alteration, Session Hijacking, System Profiling, System Hijacking, Repudiation Audit Event Management
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[OPC 10000-4](#)

Signing, GetEndpoints, SecureChannel, Audit Trail Proof of Possession

[OPC 10000-7 - ConformanceUnits](#)

[OPC 10000-7 – Profiles](#)

[User Token – JWT Server Facet](#),
[User Token – JWT Client Facet](#)
[Auditing 2022 Server Facet](#)
[Auditing Client Facet](#)
[Best Practice – Audit Events](#)

EDR 2.13: Use of physical diagnostic and test interfaces	√	√	√	N	OPC is hardware agnostic
EDR 2.13 RE (1) Active monitoring		√	√	N	OPC is hardware agnostic
HDR 2.13: Use of physical diagnostic and test interfaces	√	√	√	N	OPC is hardware agnostic
HDR 2.13 RE (1) Active monitoring		√	√	N	OPC is hardware agnostic
NDR 2.13: Use of physical diagnostic and test interfaces	√	√	√	N	OPC is hardware agnostic

NDR 2.13 RE (1) Active monitoring	√	√	N	OPC is hardware agno
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Table A.3 – ISA/IEC 62443 Mapping FR 3 System integrity

ISA/IEC 62443-4-2 CRs and Res	Related to				Applies to OPC UA	OPC UA Part #	Keyword text or comm
	SL1	SL2	SL3	SL4			
CR 3.1: Communication integrity	√	√	√	√	Y		SecureChannel – OpenSecureChannel
						OPC 10000-4	SecureChannel Service
						OPC 10000-6	SecureChannel , SecurityProtocol
						OPC 10000-7- ConformanceUnits	Security Policy Require
						OPC 10000-7 - Profiles	SecurityPolicy [A] – Aes128_Sha256_RsaOa SecurityPolicy [B] – Basic256Sha256 SecurityPolicy - Aes256 Sha256-RsaPss SecurityPolicy [A] - Pul Aes128-CTR SecurityPolicy - PubSu Aes256-CTR
CR 3.1 RE (1): Communication authentication	√	√	√	√	Y		SecureChannel – OpenSecureChannel
						OPC 10000-4	SecureChannel Service
						OPC 10000-6	SecureChannel
						OPC 10000-7- ConformanceUnits	Security Policy Require
						OPC 10000-7 - Profiles	SecurityPolicy [A] – Aes128_Sha256_RsaOa SecurityPolicy [B] – Basic256Sha256 SecurityPolicy - Aes256 Sha256-RsaPss
SAR 3.2: Protection from malicious code	√	√	√	√	N		OPC does not define requirement related to

						malicious code protection example. virus checker
EDR 3.2: Protection from malicious code	✓	✓	✓	✓	N	OPC does not define requirement related to installation or execution software
HDR 3.2: Protection from malicious code	✓	✓	✓	✓	N	OPC does not define requirement related to malicious code protection example. virus checker
HDR 3.2 RE (1): Report version of code protection		✓	✓	✓	N	OPC does not define requirement related to malicious code protection example. virus checker
NDR 3.2 – Protection from malicious code	✓	✓	✓	✓	N	OPC does not define requirement related to malicious code protection example. virus checker
CR 3.3: Security functionality verification	✓	✓	✓	✓	Y	Identity Provider, SecurityKeyService, SecureChannel , TLS
						OPC 10000-4 OpenSecureChannel, CreateSession, Write
						OPC 10000-6 OPC UA Secure Conveyance (UASC), Verifying Message Security, Token Policy, Bad_SecureChannel
						OPC 10000-7 - ConformanceUnits
						OPC 10000-7 - Profiles User Token – JWT Service Facet, User Token – JWT Client SecurityPolicy [A] – Aes128_Sha256_RsaOaep SecurityPolicy [B] – Basic256Sha256 SecurityPolicy - Aes256 Sha256-RsaPss
CR 3.3 RE (1) Security functionality			✓		N	OPC does not define function verification

verification during normal operation						
CR 3.4: Software and information integrity	√	√	√	√	P	ApplicationInstance Se Certificate OPC 10000-4 SoftwareCertificates OPC 10000-6 ApplicationInstance Se Certificate, X.509 v3 OPC 10000-7- ConformanceUnits Security ApplicationIns Security Certificate, Security Certificate Val OPC 10000-7 - Profiles Global Security Certific Management Server Global Certificate Management Client Fa
CR 3.4 RE (1): Authenticity of software and information	√	√	√		P	ApplicationInstance Se Certificate OPC 10000-4 SoftwareCertificates OPC 10000-6 ApplicationInstance Se Certificate, X.509 v3 OPC 10000-7- ConformanceUnits Security ApplicationIns Security Certificate Security Certificate Val OPC 10000-7 - Profiles Global Security Certific Management Server Global Certificate Management Client Fa
CR 3.4 RE (2) Automated notification of integrity violations		√	√		P	ApplicationInstance Se Certificate OPC 10000-4 SoftwareCertificates OPC 10000-6 ApplicationInstance Se Certificate, X.509 v3 OPC 10000-7- ConformanceUnits Security ApplicationIns Security Certificate, Security Certificate Val

						OPC 10000-7 - Profiles	Global Security Certificate Management Server Global Certificate Management Client Fa
CR 3.5: Input validation	√	√	√	√	N		OPC does not define the requirements, but does ensure that input data (method parameter or write) is of the correct datatype
CR 3.6: Deterministic output	√	√	√	√	N		OPC does not define application behaviour, does define information models that include substituted values and behaviours for failed s
CR 3.7: Error handling	√	√	√	√	Y	OPC 10000-4	Request/Response Ser
						OPC 10000-5	SessionDiagnosticsOb
						OPC 10000-6	MessageChunks, Error Handling, Error Message, CloseSecureChannel
						OPC 10000-7 - ConformanceUnits	Security Policy Require
						OPC 10000-7 - Profiles	SecurityPolicy [A] – Aes128_Sha256_RsaOa SecurityPolicy [B] – Basic256Sha256 SecurityPolicy - Aes256 Sha256-RsaPss
CR 3.8: Session integrity	√	√	√	Y			SecureChannel , Session
						OPC 10000-4	Session Service Set, Cr Session, Auditing Sess Service, SessionAuthentication
						OPC 10000-7 - ConformanceUnits	Session Base Discovery Get Endpoir
						OPC 10000-7 - Profiles	Standard UA Client 201 Profile , Base Server Behaviour

CR 3.9: Protection of audit information	√	√	√	P		OPC provides security to Audit event generation does not define Audit Logging requirements tools for analysing audit records
						SecureChannel , Session
					OPC 10000-4	Session Service Set, Create Session, Auditing Session Service, SessionAuthentication
					OPC 10000-7- ConformanceUnits	Session Base Discovery Get Endpoint
					OPC 10000-7 - Profiles	Standard UA Client 2011 Profile , Base Server Behaviour
CR 3.9 RE (1) Audit records on write-once media			√	N		OPC does not define hardware requirement
EDR 3.10: Support for updates	√	√	√	√	N	Some companion specification can define additional functionality would apply, such as OPC 10000-100
EDR 3.10: RE (1): Update authenticity and integrity	√	√	√	N		
HDR 3.10: Support for updates	√	√	√	√	N	
HDR 3.10 RE (1): Update authenticity and integrity	√	√	√	N		
NDR 3.10 – Support for updates	√	√	√	√	N	

NDR 3.10 RE (1) Update authenticity and integrity	√	√	√	N	
EDR 3.11: Physical tamper resistance and detection	√	√	√	N	
EDR 3.11 RE (1) Notification of a tampering attempt		√	√	N	
HDR 3.11: Physical tamper resistance and detection	√	√	√	N	
HDR 3.11 RE (1) Notification of a tampering attempt		√	√	N	
NDR 3.11 – Physical tamper resistance and detection	√	√	√	N	
NDR 3.11 RE (1) Notification of a tampering attempt		√	√	N	
EDR 3.12: Provisioning product supplier roots of trust	√	√	√	N	
HDR 3.12: Provisioning product supplier roots of trust	√	√	√	N	
NDR 3.12 – Provisioning product	√	√	√	N	

supplier roots of trust					
EDR 3.13: Provisioning asset owner roots of trust	√	√	√	N	
HDR 3.13: Provisioning asset owner roots of trust	√	√	√	N	
NDR 3.13 – Provisioning asset owner roots of trust	√	√	√	N	
EDR 3.14: Integrity of the boot process	√	√	√	√	N
EDR 3.14 RE (1): Authenticity of the boot process	√	√	√	√	N
HDR 3.14: Integrity of the boot process	√	√	√	√	N
HDR 3.4 RE (1): Authenticity of the boot process	√	√	√	√	N
NDR 3.14 – Integrity of the boot process	√	√	√	√	N
NDR 3.14 RE (1) Authenticity of the boot process	√	√	√	√	N

Table A.4 – ISA/IEC 62443 Mapping FR 4 Data confidentiality

ISA/IEC 62443-4-2 CRs and Res	Related to				Applies to OPC UA	OPC UA Part #	Keyword text or comment
	SL1	SL2	SL3	SL4			

CR 4.1: Information confidentiality	√	√	√	√	Y		Confidentiality, Eavesdropping, Client/Server, PubSub, Confidentiality
						OPC 10000-4	SecureChannel Service Set
						OPC 10000-6	OPC UA HTTPS, WebSockets (Security)
						OPC 10000-7- ConformanceUnits	Security Policy Required
						OPC 10000-7 - Profiles	SecurityPolicy [A] – Aes128_Sha256_RsaOae SecurityPolicy [B] – Basic256Sha256 SecurityPolicy - Aes256-Sha256-RsaPss
CR 4.2: Information persistence		√	√	√	N		OPC does not describe hardware
CR 4.2 RE (1) Erase of shared memory resources			√	√	N		OPC does not describe hardware
CR 4.2 RE (2) Erase verification			√	√	N		OPC does not describe hardware
CR 4.3: Use of cryptography	√	√	√	√	Y		Asymmetric Cryptography, Cryptography, Symmetric Cryptography, SecurityPolicies, Random Number Generation, Security Certificate Management
						OPC 10000-4	GetEndpoints, OpenSecureChannel
						OPC 10000-6	Security Handshake, Security Certificates, AccessTokens, Security

				Header, Deriving Keys (Table 49)
			OPC 10000-7- ConformanceUnits	
			OPC 10000-7 - Profiles	Best Practice – Random Numbers AccessToken Request Client Facet, Security User Access Control Base Profile, Global Discovery and Security Certificate Management 2017 Server, Global Security Certificate Management Client 2017 Profile
			OPC 10000-12	Certificate Management Overview, KeyCredential Management

Table A.5 – ISA/IEC 62443 Mapping FR 5 Restricted data flow

ISA/IEC 62443- 4-2 CRs and Res	Related to				Applies to OPC UA	OPC UA Part #	Keyword text or comment
	SL1	SL2	SL3	SL4			
CR 5.1: Network segmentation	√	√	√	√	P		OPC describe network segmentation in part 2, but does not require it or provide any additional support for it.
							Network Segmentation, OpenSecureChannel
						OPC 10000-7 ConformanceUnits	
						OPC 10000-7 - Profiles	Standard UA Client 2022 Profile, Base Server Behaviour Facet

NDR 5.2 – Zone boundary protection	√	√	√	√	N		OPC does not define network hardware.
NDR 5.2 RE (1) Deny all, permit by exception		√	√	√	N		OPC does not define network hardware.
NDR 5.2 RE (2) Island mode			√	√	N		OPC does not define network hardware.
NDR 5.2 RE (3) Fail close			√	√	N		OPC does not define network hardware.
NDR 5.3 – General purpose, person-to- person communication restrictions	√	√	√	√	N		OPC does not define network hardware.
CR 5.4 – Application partitioning	NA	NA	NA	NA	NA		Nothing defined in IEC 62443

Table A.6 – ISA/IEC 62443 Mapping FR 6 Timely response to events

ISA/IEC 62443-4-2 CRs and Res	Related to				Applies to OPC UA	OPC UA Part #	Keyword text or comment
	SL1	SL2	SL3	SL4			
CR 6.1: Audit log accessibility	√	√	√	√	N		OPC does not define an Audit log. OPC defines standard Audit records and also defined historical storage of events. It provides access restriction for all data.
CR 6.1 RE (1) Programmatic access to audit logs			√	√	N		OPC does not define an Audit log. OPC defines standard Audit records and also defined

historical storage of events. It provides access restriction for all data, standard methods for programmatical access are defined.

CR 6.2: Continuous monitoring	√	√	√	Y	OPC 10000-7- ConformanceUnits	Monitor Items, GetMonitoredItems Method, SetMonitoringMode.
					OPC 10000-7 - Profiles	Subscription Server Facet, Standard UA Client 2022 Profile , Standard DataChange Subscription 2017 Server Facet

Table A.7 – ISA/IEC 62443 Mapping FR 7 Resource availability

ISA/IEC 62443-4-2 CRs and Res	Related to				Applies to OPC UA	OPC UA Part #	Keyword text or comment
	SL1	SL2	SL3	SL4			
CR 7.1: Denial of service protection	√	√	√	√	Y		Application Crashes, Fuzz Testing, Certification
						OPC 10000-4	CreateSession, OpenSecureChannel, AuthenticationToken
						OPC 10000-7- ConformanceUnits	Session Base Discovery Get Endpoints
						OPC 10000-7 - Profiles	Standard UA Client 2022 Profile , Base Server Behaviour Facet
CR 7.1 RE (1): Manage communication load from component	√	√	√	√	Y		Message flooding, GetEndpoints, OpenSecureChannel
						OPC 10000-4	CreateSession, OpenSecureChannel,

						AuthenticationToken	
						OPC 10000-7- ConformanceUnits	Session Base Discovery Get Endpoints
						OPC 10000-7 - Profiles	Standard UA Client 2022 Profile, Base Server Behaviour Facet
CR 7.2: Resource management	√	√	√	√	Y		Resource exhaustion, ClientAuthentication, ServerAuditing, OpenSecureChannel
						OPC 10000-4	CreateSession, OpenSecureChannel, AuthenticationToken
						OPC 10000-7- ConformanceUnits	Session Base Discovery Get Endpoints
						OPC 10000-7 - Profiles	Standard UA Client 2022 Profile, Base Server Behaviour Facet
CR 7.3: Control system backup	√	√	√	√	N		OPC does not define system level backup requirements.
CR 7.3 RE (1): Backup integrity verification		√	√	√	N		OPC does not define verification of backup requirements.
CR 7.4: Control system recovery and reconstitution					N		OPC UA does not define backup and restore requirements (to a specific state).
CR 7.5 - Emergency Power	NA	NA	NA	NA	NA		(no requirements for this)
CR 7.6: Network and security	√	√	√	√	P		OPC UA requires that application can be configured for network and security

configuration settings						setting, but it does not define how this is accomplished.
						ClientAuthentication, OpenSecureChannel
						OPC 10000-4 CreateSession, OpenSecureChannel, Discovery
						OPC 10000-7 - ConformanceUnits Session Base Discovery Get Endpoints
						OPC 10000-7 - Profiles Standard UA Client 2022 Profile
CR 7.6 RE (1) Machine-readable reporting of current security settings	√	√		P		This is not defined in an OPA UA specification, but OPC UA does define a machine readable XML format that an application could use to export the security setting.
						ClientAuthentication, OpenSecureChannel
						OPC 10000-4 CreateSession, OpenSecureChannel, Discovery
						OPC 10000-7 - ConformanceUnits Session Base Discovery Get Endpoints
						OPC 10000-7 - Profiles Standard UA Client 2022 Profile
CR 7.7: Least functionality	√	√	√	√	N	The OPC UA specification does describe that least functionality is recommended with regard to OPC UA access, but this requirement applies to all service/ports

					protocols, etc. which is beyond the scope of OPC UA Specifications.
CR 7.8: Control system component inventory	√	√	√	N	This scope is beyond what is defined in core OPC UA specifications. Some companion specification can define additional details about the overall environment in which OPC UA is executing, which could be able to cover this requirement.

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